



part V

victory

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eureka!

As the computer revolution flooded the modern world with data, Bayes' rule faced one of its biggest crises in 250 years. Was an eighteenth-century theory—discovered when statistical facts were scarce and computation was slow and laborious—doomed to oblivion? It had already survived five near-fatal blows: Bayes had shelved it; Price published it but was ignored; Laplace discovered his own version but later favored his frequency theory; frequentists virtually banned it; and the military kept it secret.

By 1980 anyone studying the environment, economics, health, education, or social science was tap-tapping data into a terminal connected to a mainframe computer. “Input” became a verb. Medical records, for example, included dozens of measurements of every patient, ranging from age, gender, and race to blood pressure, weight, heart attacks, and smoking history. Wine statistics included chemical measurements and quality scores for every vintner, varietal, and vintage.

But who knew which of the 20-odd attributes of a patient or a wine were important? Researchers needed to analyze more than one unknown at once, calculate the relationships among multiple variables, and determine the effect that a change in one had on others. Yet real-life facts did not fall into tidy bell-shaped curves, and each time the variables were refined, more unknowns cropped up. Computers were generating a multivariate revolution and spawning a plague of unknowns called the curse of high-dimensionality. Statisticians had to wonder whether a method ideal for tossing a few gold coins could adapt to the new world.

Bayesians were still a small and beleaguered band of a hundred or more

in the early 1980s. Computations took forever, so most researchers were still limited to “toy” problems and trivialities. Models were not complex enough. The title of a meeting held in 1982, “Practical Bayesian Statistics,” was a laughable oxymoron. One of Lindley’s students, A. Philip Dawid of University College London, organized the session but admitted that “Bayesian computation of any complexity was still essentially impossible. . . . Whatever its philosophical credentials, a common and valid criticism of Bayesianism in those days was its sheer impracticability.”¹

The curse of dimensionality plagued both Bayesians and frequentists. Many in the academic statistical community still debated whether to indulge in computer-intensive analysis at all. Most statisticians of the era were mathematicians, and many confused their beloved old calculators—their manual Brunsvigas and electric Facits—with the new electronic computers. They tried to analyze the new data with methods designed for old calculating tools. One statistician boasted that his calculating procedure consisted of marching into his university’s computer center and saying, “Get on with it.”² Thanks to pioneers like Robert Schlaifer and Howard Raiffa, Bayesians held sway in business schools and theoretical economics, while statistics departments were dominated by frequentists, who focused on data sets with few unknowns rather than on those packed with unknowns.

As a result, many statistics departments watched from the sidelines as physical and biological scientists analyzed data about plate tectonics, pulsars, evolutionary biology, pollution, the environment, economics, health, education, and social science. Soon engineers, econometricians, computer scientists, and information technologists acquired the cachet that humdrum statisticians seemed to lack. Critics sniffed that statistics departments were isolated, defensive, and on the decline. Leading statistical journals were said to be so mathematical that few could read them and so impractical that few would want to. The younger generation seemed to think that computers and their algorithms could replace mathematics entirely.

In what could have been a computational breakthrough, Lindley and his student Adrian F. M. Smith showed Bayesians how to develop models by breaking complex scientific processes into stages called hierarchies. The system would later become a Bayesian workhorse, but at the time it fell flat on its face. The models were too specialized and stylized for many scientific applications. It would be another 20 years before Bayesian textbooks taught hierarchical models. Mainstream statisticians and scientists simply did not believe that Bayes could ever be practical. Indicative of their attitude is the

fact that while Thomas Bayes' clerical ancestors were listed in Britain's *Dictionary of National Biography* he himself was not.

Yet amazingly, amid these academic doubts, a U.S. Air Force contractor used Bayes to analyze the risk of a *Challenger* space shuttle accident. The air force had sponsored Albert Madansky's Bayesian study at the RAND Corporation during the Cold War, but the National Aeronautics and Space Administration (NASA) still distrusted subjective representations of uncertainty. Consequently, it was the air force that sponsored a review in 1983 of NASA's estimates of the probability of a shuttle failure. The contractor, Teledyne Energy Systems, employed a Bayesian analysis using the prior experience of 32 confirmed failures during 1,902 rocket motor launches. Using "subjective probabilities and operating experience," Teledyne estimated the probability of a rocket booster failure at 1 in 35; NASA's estimate at the time was 1 in 100,000. Teledyne, however, insisted that "the prudent approach is to rely on conservative failure estimates based on prior experience and probabilistic analysis."³ On January 28, 1986, during the shuttle's twenty-fifth launch, the *Challenger* exploded, killing all seven crew members aboard.

The disparity between the military's sometime acceptance of Bayes and the academic statistical community's refusal to embrace it is still puzzling. Did the military's top-secret experience with Bayes during the Second World War and the Cold War give it confidence in the method? Was the military less afraid of using computers? Or did it simply have easier access to powerful ones? Given that many sources dealing with the Second World War and the Cold War are still classified, we may never know the answers to these questions.

Several civilian researchers tackling hitherto intractable problems concerning public health, sociology, epidemiology, and image restoration did experiment during the 1980s with computers for Bayes. A major controversy about the effect of diesel engine emissions on air quality and cancer inspired the first attempt. By the 1980s cancer specialists had solid data about the effects of cigarette smoke on people, laboratory animals, and cells but little accurate information about diesel fumes. William H. DuMouchel from MIT's mathematics department and Jeffrey E. Harris from its economics department and Massachusetts General Hospital teamed up in 1983 to ask, "Could you borrow and extrapolate and take advantage of information from non-human species for humans?"⁴ Such meta-analyses, combining the results of similar trials, were too complex for frequentists to address, but DuMouchel was a disciple of Smith and his hierarchical work with Lindley. Harris was not a

statistician and did not care what method he used as long as it answered the question. Adopting hierarchical Bayes, they borrowed information from laboratory tests on mice, hamster embryo cells, and chemical substances. They even incorporated experts' opinions about the biological relevance of non-humans to humans and of cigarette to diesel smoke. Bayes let them account formally for their uncertainties about combining information across species.

Microcomputers were not widely available. Many of the researchers studying the new acquired immune deficiency syndrome (AIDS) epidemic, for example, were making statistical calculations by hand, and mathematical shortcuts were still being published for them. Harris programmed the diesel project in APL, a language used for matrix multiplications, and sent it via teletype to MIT's computer center. He drew illustrations on poster boards, added captions by pressing on wax letters, and arranged for an MIT photographer to take their pictures.

Thanks to mice and hamster studies, DuMouchel and Harris were able to conclude that even if light-duty diesel vehicles captured a 25% market share over 20 years, the risk of lung cancer would be negligible for the typical urban resident compared to the typical pack-a-day cigarette smoker. The smoker's risk was 420,000 times worse. Today, Bayesian meta-analyses are statistically old hat, but DuMouchel and Harris made Bayesians salivate for more big-data methods—and for the computing power to deal with them.

While lung cancer researchers explored Bayes, Adrian Raftery was working at Trinity College in Dublin on a well-known set of statistics about fatal coal-dust explosions in nineteenth-century British mines. Previous researchers had used frequency techniques to show that coal mining accident rates had changed over time. They assumed, however, that the change had been gradual. Raftery wanted to check whether it had been gradual or abrupt. First, he developed some heavy frequentist mathematics for analyzing the data. Then, out of curiosity, he experimented with Bayes' rule, comparing a variety of theoretical models to see which had the highest probability of determining when the accidents rates actually changed. "I found it very easy. I just solved it very, very quickly," Raftery recalled. And in doing so he discovered a remarkable, hitherto unknown event in British history. Raftery's Bayesian analysis revealed that accident rates plummeted suddenly in the late 1880s or early 1890s. A historian friend suggested why. In 1889, British miners had formed the militant Miners' Federation (which later became the National Union of Mine Workers). Safety was their number one issue. Almost overnight, coal mines got safer.

“It was a Eureka moment,” Raftery said. “It was quite a thrill. And without Bayesian statistics, it would have been much harder to do a test of this hypothesis.”⁵ Frequency-based statistics worked well when one hypothesis was a special case of the other and both assumed gradual behavior. But when hypotheses were competing and neither was a special case of the other, frequentism was not as helpful, especially with data involving abrupt changes—like the formation of a militant union.

Raftery wound up publishing two papers in 1986 about modeling abrupt rate changes. His first, frequentist paper was long, dense, and virtually unread. His second, Bayesian paper was shorter, simpler, and had a much greater impact. Raftery’s third 1986 paper ran just 1-1/4 pages and had an immediate effect on sociologists. The article appeared just as many sociologists were about to give up on frequentism’s controversial *p*-values. A typical sociologist might work with data sets about thousands of individuals, each with hundreds of variables such as age, race, religion, climate, and family structure. Unfortunately, when researchers tried to determine the relevance of those variables using frequentist methods developed by Karl Pearson and R. A. Fisher for 50 to 200 cases, the results were often bizarre. Obscure effects became important, or went in opposite directions, or were disproved by later studies. By selecting a single model for large samples, frequentists ignored uncertainties about the model. Yet few social scientists could repeat their surveys or rerun experiments under precisely the same conditions. By the early 1980s many sociologists had concluded that, for testing hypotheses, their intuition was more accurate than frequentism.

Bayes, on the other hand, seemed to produce results that corresponded more closely to sociologists’ intuition. Raftery told his colleagues, “The point is that we should be comparing the models, not just looking for possibly minor discrepancies between one of them and the data.”⁶ Researchers really want to know which of their models is more likely to be true, given the data. With Bayes, researchers could study sudden shifts from one stable form to another in biological growth phases, trade deficits and economic behavior, the abandonment and resettlement of archaeological sites, and clinical conditions such as rejection and recovery in organ transplantation and brain waves in Parkinson’s disease. Bayesian hypothesis testing swept sociology and demography, and Raftery’s short paper is still among the most cited in sociology.

Meanwhile, image processing and analysis had become critically important for the military, industrial automation, and medical diagnosis. Blurry, distorted, imperfect images were coming from military aircraft, infrared

sensors, ultrasound machines, photon emission tomography, magnetic resonance imaging (MRI) machines, electron micrographs, and astronomical telescopes. All these images needed signal processing, noise removal, and deblurring to make them recognizable. All were inverse problems ripe for Bayesian analysis.

The first known attempt to use Bayes to process and restore images involved nuclear weapons testing at Los Alamos National Laboratory. Bobby R. Hunt suggested Bayes to the laboratory and used it in 1973 and 1974. The work was classified, but during this period he and Harry C. Andrews wrote a book, *Digital Image Restoration*, about the basic methodology; the laboratory declassified the book and approved its publication in 1976. The U.S. Congress retained Hunt in 1977 and 1978 to analyze images of the shooting of President Kennedy. In his testimony, Hunt did not refer to Bayes. "Too technical for a Congressional hearing," he said later.

At almost the same time that Hunt was working on image analysis for the military, Julian Besag at the University of Durham in England was using diseased tomato plants to study the spread of epidemics. Bayes helped him discern local regularities and neighborly interactions among plants growing in pixel-like lattice systems. Looking at one pixel, Besag realized he could estimate the probability that its neighbor might share the same color, a useful tool for image enhancement. But Besag was not a card-carrying Bayesian, and his work went largely unnoticed at the time.

A group of researchers with Ulf Grenander at Brown University was trying to design mathematical models for medical imaging by exploring the effect one pixel could have on a few of its neighbors. The calculations involved easily a million unknowns. Grenander thought that once Bayes was embedded in a realistic problem, philosophical objections to it would fade.

Stuart Geman was attending Grenander's seminar in pattern theory, and he and his brother Donald Geman tried restoring a blurry photograph of a roadside sign. The Geman brothers were interested in noise reduction and in finding ways to capture and exploit regularities to sharpen the lines and edges of unfocused images. Stuart had majored in physics as an undergraduate and knew about Monte Carlo sampling techniques. So the Geman brothers invented a variant of Monte Carlo that was particularly suited to imaging problems with lots of pixels and lattices.

Sitting at a table in Paris, Donald Geman thought about naming their system. A popular Mother's Day gift at the time was a Whitman's Sampler assortment of chocolate bonbons; a diagram inside the box top identified

the filling hidden inside each candy. To Geman, the diagram was a matrix of unknown but enticing variables. "Let's call it Gibbs sampler," he said, after Josiah Willard Gibbs, a nineteenth-century American physicist who applied statistical methods to physical systems.⁷

The dots were starting to connect. But the Gemans, like Besag, operated in a small niche field, spatial statistics. And instead of nibbling at their problem a pixel at a time, the Gemans tried gobbling it whole. Working at pixel levels on a 64 x 64-cell fragment of a photo, they produced too many unknowns for computers of the day to digest. They wrote up their sampler in a formidably difficult paper and published it in 1984 in *IEEE Transactions on Pattern Analysis and Machine Intelligence*. Specialists in image processing, neural networks, and expert systems quickly adopted the method, which, with computers gaining more power every year, also sparked the interest of some statisticians. The brothers spent the next year racing around the globe giving invited talks.

Donald Geman used the Gibbs sampler to improve satellite images; Stuart used it for medical scans. Several years later, statisticians outside the small spatial imaging community began to realize that more general versions could be useful. The Gibbs sampler's flexibility and reliability would make it the most popular Monte Carlo algorithm. Still later the West learned that a Russian dissident mathematician, Valentin Fedorovich Turchin, had discovered the Gibbs sampler in 1971, but his work had been published in Russian-language journals, did not involve computers, and was overlooked.

By 1985 the old argument between Bayesians and frequentists was losing its polarizing zing, and Glenn Shafer of Rutgers University thought it had "calcified into a sterile, well-rehearsed argument." Persi Diaconis made a similar but nonetheless startling observation, one that no one familiar with the battles between Bayesians, Karl Pearson, Ronald Fisher, and Jerzy Neyman could have imagined. "It's nice that our field is so noncompetitive," Diaconis said. "If you take many other fields, like biology, people just slice each other up."⁸

Still, the conviction remained that without more powerful and accessible computers and without user-friendly and economical software, computing realistic problems with Bayes was impossible.

Lindley had been programming his own computers since 1965 and regarded Bayes as ideal for computing: "One just feeds in the axioms and the data and allows the computer to follow the laws of arithmetic." He called it "turning the Bayesian crank." But his student Smith saw something the

older man did not: the key to making Bayes useful in the workplace would be computational ease, not more polished theory. Later Lindley wrote, “I consider it a major mistake of my professional life, not to have appreciated the need for computing rather than mathematical analysis.”⁹

Ignoring the defensive posture of many statistics departments, Smith launched an offensive in a radically new direction. Smith’s friends think of him as a lively, practical man with street smarts, people skills, and a can-do personality, the kind of person more comfortable in running shorts than academic garb. He was certainly willing to do the dirty work needed to make Bayes practical. He learned Italian to help translate de Finetti’s two-volume *Theory of Probability* and then got it published. For the first time de Finetti’s subjectivist approach was widely available to Anglo-American statisticians. Smith also developed filters, practical computer devices that would ease Bayesian computations enormously.

Next, Smith and three others—Lindley, José M. Bernardo, and Morris DeGroot—organized an international conference series for Bayesians in Valencia, Spain. It has been held regularly since 1979. Smith expected “the usual criticism from non-Bayesians in reaction to whatever I say.” Sure enough, frequentists accused Bayesians of sectarian habits, meetings in remote locations, and mock cabarets featuring skits and songs with Bayesian themes. Other disciplines have done the same, of course. The conferences played a vital role in helping to build camaraderie in a small field under attack.

In 1984 Smith issued a manifesto—and italicized it for emphasis: “Efficient numerical integration procedures are the key to more widespread use of Bayesian methods.”¹⁰ With computerized data collection and storage, hand analyses were becoming impossible. When microcomputers appeared, attached to fast networks with graphics and vast storage capabilities, data analysts could finally hope to improvise as easily as they had with pencil and paper. With characteristic practicality, Smith set his University of Nottingham students to work developing efficient, user-friendly software for Bayesian problems in spatial statistics and epidemiology.

Intrigued with Smith’s projects, Alan Gelfand at the University of Connecticut asked Smith if he could spend his sabbatical year at Nottingham. When Gelfand arrived, Smith suggested they start something new. Gelfand recalled, “He gave me this Tanner and Wong paper, saying, ‘This is kind of an interesting paper. There must be something more to it.’” Wing Hung Wong and Martin A. Tanner, at the Universities of Chicago and Wisconsin, respectively, were interested in spatial image analysis for identifying genetic

linkages and for scanning the brain using positron emission tomography (PET). Wong had been adapting for Bayesians the EM algorithm, an iterative system secretly developed by the National Security Agency during the Second World War or the early Cold War. Arthur Dempster and his student Nan Laird at Harvard discovered EM independently a generation later and published it for civilian use in 1977. Like the Gibbs sampler, the EM algorithm worked iteratively to turn a small data sample into estimates likely to be true for an entire population.

Gelfand was studying the Wong paper when David Clayton from the University of Leicester dropped by and said, "Oh, the paper by Geman and Geman has something to do with it, I think." Clayton had written a technical report which, although never published, concerned Gibbs sampling. The minute Gelfand saw the Gemans' paper, the pieces came together: Bayes, Gibbs sampling, Markov chains, and iterations. A Markov chain can be composed of scores of links, and for each one a range of possible variables must be sampled and calculated one after another. Anyone studying a rare and subtle effect must compute each chain over and over again to get a big enough number to reveal the rarity. The numbers involved become enormous, and the length and tediousness of the calculations turned off most researchers.

But Gelfand and Smith saw that replacing difficult integration with sampling would be a wonderful computational tool for Bayesians. "It went back to the most basic things you learn in an introductory statistics course," Gelfand explained. "If you want to learn about a distribution or a population, you take samples from it. But don't sample directly." Imaging and spatial statisticians had been looking at local models as a whole, but Gelfand and Smith realized they should build long chains, series of observations generated one or two at a time, one after another. As Gelfand explained, "The trick was to look at simple distributions one at a time but never look at the whole. The value of each one depended only on the preceding value. Break the problem into tiny pieces that are easy to solve and then do millions of iterations. So you replace one high-dimensional draw with lots of low-dimensional draws that are easy. The technology was already in place. That's how you break the curse of high-dimensionality."¹¹

Smith and Gelfand wrote their article as fast as they could. The elements of their system were already known, but their grand synthesis was new. Once others thought about it, they'd realize the importance of the method too.

When Smith spoke at a workshop in Quebec in June 1989, he showed that Markov chain Monte Carlo could be applied to almost any statistical

problem. It was a revelation. Bayesians went into “shock induced by the sheer breadth of the method.”¹² By replacing integration with Markov chains, they could finally, after 250 years, calculate realistic priors and likelihood functions and do the difficult calculations needed to get posterior probabilities.

To outsiders, one of the amazing aspects of Bayes’ history is that physicists and statisticians had known about Markov chains for decades. To illustrate this puzzling lapse, some flashbacks are required. Monte Carlo began in 1906, when Andrei Andreyevich Markov, a Russian mathematician, invented Markov chains of variables. The calculations took so long, though, that Markov himself applied his chains only to the vowels and consonants in a Pushkin poem.

Thirty years later, with the beginning of nuclear physics in the 1930s, the Italian physicist Enrico Fermi was studying neutrons in collision reactions. Fermi fought insomnia by computing Markov chains in his head to describe the paths of neutrons in collision reactions. To the surprise of his colleagues the next morning, Fermi could predict their experimental results. With a small mechanical adding machine, Fermi built Markov chains to solve other problems too. Physicists called the chains statistical sampling.

Fermi did not publish his methods, and, according to Jack Good, government censorship kept Markov chains under tight wraps during the Second World War. After the war Fermi helped John von Neumann and Stanislaw Ulam develop the technique for hydrogen-bomb developers using the new ENIAC computer at the University of Pennsylvania. To calculate the critical mass of neutrons needed to make a thermonuclear explosion, Maria Goepfert Mayer, a future Nobel Prize-winning physicist, simulated the process with Markov chains, following one neutron at a time and making decisions at various places as to whether the neutron was most likely to get absorbed, escape, die, or fission. The calculation was too complicated for existing IBM equipment, and many thought the job was also beyond computers. But Mayer reported that it did “not strain the capacity of the Eniac.”¹³ In 1949 she spoke at a symposium organized by the National Bureau of Standards, Oak Ridge National Laboratory, and RAND for physicists to brief mathematicians and statisticians on hitherto classified applications.

That same year Nicholas Metropolis, who had named the algorithm Monte Carlo for Ulam’s gambling uncle, described the method in general terms for statisticians in the prestigious *Journal of the American Statistical Association*. But he did not detail the algorithm’s modern form until 1953, when his article appeared in the *Journal of Chemical Physics*, which is generally found only in

physics and chemistry libraries. Moreover, he and his coauthors—two husband-and-wife teams, Arianna and Marshall Rosenbluth and Augusta and Edward Teller—were concerned strictly with particles moving around a square. They did not generalize the method for other applications. Thus it was physicists and chemists who pioneered Monte Carlo methods. Working on early computers with between 400 and 80,000 bytes of memory, they dealt with memory losses, unreadable tapes, failed vacuum tubes, and programming in assembly language. Once it took literally months to track down a small programming error. In the 1950s RAND developed a lecture series on Monte Carlo techniques and used them in a specially built simulation laboratory to test case after case of problems too complex for mathematical formulas.

During this difficult period, statisticians were advised several more times to use Monte Carlo either with or without computers. In 1954 two statisticians associated with the British Atomic Energy Research Establishment recommended that readers of the *Journal of the Royal Statistical Society* consult “Poor Man’s Monte Carlo” for pen-and-paper calculations; John M. Hammersley and Keith W. Morton said Monte Carlo was as easy as “simple knitting.” Lindley described Markov chains in 1965 in a text for college students.

In a particularly poignant case, W. Keith Hastings, a mathematician at the University of Toronto, was approached by chemists who were studying 100 particles interacting with each other while subject to outside forces. Because of the 600 variables in the case, Hastings said he immediately realized the importance of Markov chains for mainstream statistics and devoted all his time to them: “I was excited because the underlying idea goes way back to Metropolis. As soon as I realized it, it was off to the races. I just had to work on it. I had no choice.” In 1970 he published a paper generalizing Metropolis’s algorithm in the statistical journal *Biometrika*. Again, Bayesians ignored the paper. Today, computers routinely use the Hastings–Metropolis algorithm to work on problems involving more than 500,000 hypotheses and thousands of parallel inference problems.

Hastings was 20 years ahead of his time. Had he published his paper when powerful computers were widely available, his career would have been very different. As he recalled, “A lot of statisticians were not oriented toward computing. They take these theoretical courses, crank out theoretical papers, and some of them want an exact answer.”¹⁴ The Hastings–Metropolis algorithm provides estimates, not precise numbers. Hastings dropped out of research and settled at the University of Victoria in British Columbia in 1971. He learned about the importance of his work after his retirement in 1992.

Why did it take statisticians so long to understand the implications of an old method? And why were Gelfand and Smith first? “The best thing I can say for us is that we just sort of stumbled on it. We got lucky,” Gelfand says today. “It was just sort of sitting there, waiting for people to put the pieces together.”

Timing was important too. Gelfand and Smith published their synthesis just as cheap, high-speed desktop computers finally became powerful enough to house large software packages that could explore relationships between different variables. Bayes was beginning to look like a theory in want of a computer. The computations that had irritated Laplace in the 1780s and that frequentists avoided with their variable-scarce data sets seemed to be the problem—not the theory itself.

Yet Smith and Gelfand still thought of Monte Carlo as a last resort to be used in desperation for complicated cases. They wrote diffidently, careful to use the B-word only five times in 12 pages. “There was always some concern about using the B-word, a natural defensiveness on the part of Bayesians in terms of rocking the boat,” Gelfand said. “We were always an oppressed minority, trying to get some recognition. And even if we thought we were doing things the right way, we were only a small component of the statistical community and we didn’t have much outreach into the scientific community.”¹⁵

The Gelfand–Smith paper was an “epiphany in the world of statistics,” as Bayesians Christian P. Robert and George Casella reported. And just in case anyone missed their point, they added: “Definition: epiphany *n*. A spiritual event . . . a sudden flash of recognition.” Years later, they still described its impact in terms of “sparks,” “flash,” “shock,” “impact,” and “explosion.”¹⁶

Shedding their diffidence, Gelfand and Smith wrote a second paper six months later with Susan E. Hills and Amy Racine-Poon. This time they punctuated their mathematics exuberantly with words like “surprising,” “universality,” “versatility,” and “trivially implemented.” They concluded grandly, “The potential of the methodology is enormous, rendering straightforward the analysis of a number of problems hitherto regarded as intractable from a Bayesian perspective.”¹⁷ Luke Tierney at Carnegie Mellon tied the technique to Metropolis’s method, and the entire process—which physicists had called Monte Carlo—was baptized anew as Markov chain Monte Carlo, or MCMC for short. The combination of Bayes and MCMC has been called “arguably the most powerful mechanism ever created for processing data and knowledge.”¹⁸

When Gelfand and Smith gave an MCMC workshop at Ohio State University early in 1991, they were astonished when almost 80 scientists appeared.

They weren't statisticians, but they had been using Monte Carlo in archaeology, genetics, economics, and other subjects for years.

The next five years raced by in a frenzy of excitement. Problems that had been nightmares cracked open as easily as eggs for an omelet. A dozen years earlier the conference title "Practical Bayesian Statistics" had been a joke. But after 1990 Bayesian statisticians could study data sets in genomics or climatology and make models far bigger than physicists could ever have imagined when they first developed Monte Carlo methods. For the first time, Bayesians did not have to oversimplify "toy" assumptions.

Over the next decade, the most heavily cited paper in the mathematical sciences was a study of practical Bayesian applications in genetics, sports, ecology, sociology, and psychology. The number of publications using MCMC increased exponentially.

Almost instantaneously MCMC and Gibbs sampling changed statisticians' entire method of attacking problems. In the words of Thomas Kuhn, it was a paradigm shift.¹⁹ MCMC solved real problems, used computer algorithms instead of theorems, and led statisticians and scientists into a world where "exact" meant "simulated" and repetitive computer operations replaced mathematical equations. It was a quantum leap in statistics.

When Smith and Gelfand published their paper, frequentists could do a vast amount more than Bayesians. But within years Bayesians could do more than frequentists. In the excitement that followed, Stanford statistician Jun S. Liu, working with biologist Charles E. Lawrence, showed genome analysts that Bayes and MCMC could reveal motifs in protein and DNA. The international project to sequence the human genome, launched in 1990, was generating enormous amounts of data. Liu showed how a few seconds on a workstation programmed for Bayes and iterative MCMC sampling could detect subtle, closely related patterns in protein and nucleic acid sequences. Then he and Lawrence could infer critical missing data pointing to common ancestors, structures, and functions. Soon genomics and computational biology were so overrun with researchers that Gelfand decided to look elsewhere for another research project.

Between 1995 and 2000 Bayesians developed particle filters like the Kalman filter and real-time applications in finance, image analysis, signal processing, and artificial intelligence. The number of attendees at Bayesian conferences in Valencia quadrupled in 20 years. In 1993, more than two centuries after his death, Thomas Bayes finally joined his clerical relatives in the *Dictionary of National Biography*.

Amid the Bayesian community's frenzy over MCMC and Gibbs sampling, a generic software program moved Bayesian ideas out into the scientific and computer world.

In an example of serendipity, two groups 80 miles apart worked independently during the late 1980s on different aspects of the same problem. While Smith and Gelfand were developing the theory for MCMC in Nottingham, Smith's student, David Spiegelhalter, was working in Cambridge at the Medical Research Council's biostatistics unit. He had a rather different point of view about using Bayes for computer simulations. Statisticians had never considered producing software for others to be part of their jobs. But Spiegelhalter, influenced by computer science and artificial intelligence, decided it was part of his. In 1989 he started developing a generic software program for anyone who wanted to use graphical models for simulations. Once again, Clayton was an important influence. Spiegelhalter unveiled his free, off-the-shelf BUGS program (short for Bayesian Statistics Using Gibbs Sampling) in 1991.

BUGS caused the biggest single jump in Bayesian popularity. It is still the most popular software for Bayesian analyses, and it spread Bayesian methods around the world.

"It wasn't a very big project," Spiegelhalter admits. "It was a staggeringly basic and powerful idea, relating Gibbs sampling to a graph to write generic programs."²⁰ Its simple code remains almost exactly the same today as it was in 1991.

Ecologists, sociologists, and geologists quickly adopted BUGS and its variants, WinBUGS for Microsoft users, LinBUGS for Linux, and OpenBUGS. But computer science, machine learning, and artificial intelligence also joyfully swallowed up BUGS. Since then it has been applied to disease mapping, pharmacometrics, ecology, health economics, genetics, archaeology, psychometrics, coastal engineering, educational performance, behavioral studies, econometrics, automated music transcription, sports modeling, fisheries stock assessment, and actuarial science. A Bayesian visiting a marine laboratory was surprised to discover all its scientists using BUGS with nary a statistician in sight.

Medical research and diagnostic testing were among the earliest beneficiaries of Bayes' new popularity. Just as the MCMC frenzy appeared to be moderating, Peter Green of Bristol University showed Bayesians how to compare the elaborate hypotheses that scientists call models. Before 1996 anyone making

a prediction about the risk of a stroke had to focus on one model at a time. Green showed how to jump between models without spending an infinite time on each. Previous studies had identified 10 possible factors involved in strokes. Green identified the top four: systolic blood pressure, exercise, diabetes, and daily aspirin.

Medical testing, in particular, benefited from Bayesian analysis. Many medical tests involve imaging, and Larry Bretthorst, a student of the Bayesian physicist Ed Jaynes, improved nuclear magnetic resonancing, or NMR, signal detection by several orders of magnitude in 1990. Bretthorst had studied imaging problems to improve the detection of radar signals for the Army Missile Command.

In 1991 a public frightened about the AIDS epidemic demanded universal screening for human immunodeficiency virus (HIV). Biostatisticians quickly used Bayes to demonstrate that screening the entire population for a rare disease would be counterproductive. Wesley O. Johnson and Joseph L. Gastwirth showed that a sensitive test like that for HIV virus would tell many patients they were infected with HIV when in fact they were not. The media publicized several suicides of people who had received a positive HIV test result but not realized they did not necessarily have the virus. Scaring healthy people and retesting them with more sophisticated procedures would have been extremely costly.

In much the same way, but more controversially, a Bayesian approach showed that an expensive MRI test for breast cancer might be appropriate for a woman whose family had many breast cancer patients but inappropriate for giving every woman between 40 and 50 years of age. A woman who has a mammogram every year for 10 years can be almost 100% sure of getting one false positive test result, and the resulting biopsy can cost \$1,000 to \$2,000. In the case of prostate cancer, the screening test for high blood levels of prostate-specific antigen (PSA) is highly accurate when it comes to identifying men with the cancer. Yet the disease is so rare that almost everyone who gets a positive test result is found not to have the cancer at all. (See appendix B for how to calculate a Bayesian problem involving breast cancer.)

On the other hand, Bayes also showed that people with negative test results for breast and prostate cancer cannot feel carefree. The PSA test is so insensitive that good news provides almost no assurance that a man does not actually have prostate cancer. The same is true to a lesser extent of mammography: its sensitivity is about 85 to 90%, meaning that a woman who finds a lump only a few months after getting a negative mammogram

should still see a doctor immediately. A strict Bayesian gives patients their probabilities for cancer instead of a categorical yes or no.

Because genetics involves extremely rare diseases, imperfect tests, and complicated problems where tiny errors in the data or calculations can affect decisions, Bayesian probabilities are expected to become increasingly important for diagnostic test assessment.

Spiegelhalter spent more than 10 years trying to sell the medical community on BUGS as the mathematical way to learn from experience. He argued that “advances in health-care typically happen through incremental gains in knowledge rather than paradigm-shifting breakthroughs, and so this domain appears particularly amenable to a Bayesian perspective.” He contended that “standard statistical methods are designed for summarizing the evidence from single studies or pooling evidence from similar studies, and have difficulties dealing with the pervading complexity of multiple sources of evidence.”²¹ While frequentists can ask only certain questions, a Bayesian can frame any question.

With the introduction of high-performance workstations in the 1980s it became possible to use Bayesian networks to handle medicine’s many interdependent variables, such as the fact that a patient with a high temperature will usually also have an elevated white blood count. Bayesian networks are graphs of nodes with links revealing cause-and-effect relationships. The “nets” search for particular patterns, assign probabilities to parts of the pattern, and update those probabilities using Bayes’ theorem. A number of people helped develop Bayesian networks, which were popularized in 1988 in a book by Judea Pearl, a computer scientist at UCLA. By treating cause and effect as a quantifiable Bayesian belief, Pearl helped revive the field of artificial intelligence.

Ron Howard, who had become interested in Bayes while at Harvard, was working on Bayesian networks in Stanford’s economic engineering department. A medical student, David E. Heckerman, became interested too and for his Ph.D. dissertation wrote a program to help pathologists diagnose lymph node diseases. Computerized diagnostics had been tried but abandoned decades earlier. Heckerman’s Ph.D. in bioinformatics concerned medicine, but his software won a prestigious national award in 1990 from the Association for Computing Machinery, the professional organization for computing. Two years later, Heckerman went to Microsoft to work on Bayesian networks.

The Federal Drug Administration (FDA) allows the manufacturers of medical devices to use Bayes in their final applications for FDA approval. Devices include almost any medical item that is not a drug or biological product,

items such as latex gloves, intraocular lenses, breast implants, thermometers, home AIDS kits, and artificial hips and hearts. Because they are usually applied locally and improved a step at a time, new models of a device should come equipped with objective prior information.

Pharmaceuticals are different. Unlike devices, pharmaceuticals are generally one-step, systemic discoveries so that potentially an industry could subjectively bias Bayes' prior hunch. Thus the FDA has long resisted pressure from pharmaceutical companies that want to use Bayes when applying for approval to sell a drug in the United States.

According to Spiegelhalter, however, the same battle seems to have subsided in England. Drug companies use WinBUGS extensively when submitting their pharmaceuticals for reimbursement by the English National Health Service. The process is, in Spiegelhalter's words, "very Bayesian without using the B-word" because it uses judgmental priors about a drug's cost effectiveness. International guidelines also allow Bayesian drug applications, but these guidelines are widely considered too vague to be effective.

Outside of diagnostic and medical device testing, Bayes' mathematical procedures have had little impact on basic clinical research or practice. Working doctors have always practiced an intuitive, nonmathematical form of Bayes for diagnosing patients. The biggest unknown in medicine, after all, is the question, What is causing the patient's symptoms? But traditional textbooks were organized by disease. They said that someone with, for instance, measles probably has red spots. But the doctor with a speckled patient wanted to know the inverse: the probability that the patient with red spots has measles. Simple Bayesian problems—for example, What is the probability that an exercise echocardiogram will predict heart disease?—started appearing on physicians' licensing examinations in 1992.

One of the few times physicians make even rough Bayesian calculations occurs when a patient has symptoms that could involve a life-threatening heart attack, a deep venous thrombosis, or a pulmonary embolism. To assess the danger, a physician assigns points for each of the patient's risk factors and adds the points. In the heart attack algorithm, the score determines the probability that within the next two weeks the patient will die, have a heart attack, or need coronary arteries opened. The points for thrombosis and embolism tell whether a patient has a low, medium, or high risk of developing a clot and which test can best produce a diagnosis. It is expected that software will be available soon to automatically tell doctors and patients the effect of a particular test result on a diagnosis.

Outside of medicine, endangered populations of ocean fish, whales, and other mammals were among the first to benefit from Bayes' new computational heft. Despite the U.S. Marine Mammal Protection Act of 1972, only a few visible and highly publicized species of whales, dolphins, and other marine mammals had been protected. Some exploited populations, including several whale species in the Antarctic, collapsed while being "managed." Having strong, abundant information about a species, frequentists and Bayesians could reach similar decisions, but when evidence was weak—as it often is in the case of marine mammals—only Bayes incorporated uncertainties about the data at hand and showed clearly when more information was needed.

Most whale populations rebounded during the 1980s, but in 1993 two government biologists, Barbara L. Taylor and Timothy Gerrodette, wrote, "At least part of the blame for the spectacular [past] overexploitation of the great whales can be placed on scientists being unable to agree . . . [on a] clear way to treat uncertainty. . . . In certain circumstances, a population might go extinct before a significant decline could be detected."²² During the administration of Bill Clinton, the Wildlife Protection Act was amended to accept Bayesian analyses alerting conservationists early to the need for more data.

Scientists advising the International Whaling Commission were particularly worried about the uncertainty of their measurements. Each year the commission establishes the number of endangered bowhead whales Eskimos can hunt in Arctic seas. To ensure the long-term survival of the bowheads, scientists compute 2 numbers each year: the number of bowheads and their rate of increase. The whales, perhaps the longest-lived mammals on Earth, can grow to more than 60 feet in length, weigh more than 60 tons, and eat 2 tons of food a day. They spend only about 5% of their time on the ocean surface because they can submerge for 30 minutes at a time, and they use their enormous heads to ram through ice when they need to surface for air. In spring, teams of scientists stood on tall perches to spot bowheads rounding Point Barrow, Alaska, on their annual migration into the western Arctic. The count was fraught with uncertainties.

Scientists representing an entire spectrum of opinion, from Greenpeace to whaling nations, worried that a lack of trustworthy data on bowhead populations was opening the species to too much risk. During a weeklong meeting to discuss the problem in 1991, their chair asked, "What can we do?"²³ There was complete silence. The scientists were the world's leading bowhead experts, but none of them could answer the question.

When Judith Zeh, the committee chair, got back to the department

of statistics at the University of Washington in Seattle, she talked with Raftery, who had recently moved there from Dublin. Not surprisingly, after his experience analyzing coal mining accidents, Raftery thought Bayes might help. Using it, the committee could assign uncertainties to all their data and augment visual sightings with recordings of whales vocalizing near underwater hydrophones.

Providentially, the spring of 1993 was a rewarding year for bowhead counting, and sightings plus vocalizing showed that the whales were almost assuredly increasing at a healthy rate. Their recovery indicated that protecting other great whale populations from commercial whaling might help them recover too.

The entire process—involving rival Bayesian and frequentist methods and whaling factions that often profoundly disagreed—could have been wildly contentious. But times were changing. Pragmatism ruled. Making full-scale Bayesian analyses to combine visual and acoustical data was expensive, and thus, because they confirmed previous frequentist studies, they were discontinued. Raftery moved on to using Bayes for 48-hour weather forecasting.

Other wildlife researchers picked up the Bayesian banner. When Paul R. Wade decided in 1988 to use Bayes for his Ph.D. thesis, he said, “I was off in this small area of marine mammal biology but I felt as if I were in the center of a revolution in science.” Ten years later, at the National Oceanic and Atmospheric Administration, he was comparing frequentist and Bayesian analyses of a small, isolated population of 200 or 300 beluga whales in the Arctic and Sub-Arctic waters of Cook Inlet, Alaska. The legal take by native hunters was roughly 87 whales a year. Frequentist methods would have required seven years of data collection to assess whether this catch was sustainable. With Bayes, five years of data showed that the beluga population was almost certainly declining substantially, and the experiment could stop. “With a small population, even a two-year delay can be important,” Wade said.²⁴ In May 1999 a hunting moratorium went into effect for the Cook Inlet belugas.

Meanwhile, a committee of the National Research Council in the National Academy of Sciences strongly recommended the aggressive use of Bayesian methods to improve estimates of marine fish stocks too. Committee members emphasized in 1998 that, because the oceans are vast and opaque, wildlife managers need realistic measurements of the uncertainties in their observations and models. Otherwise, policymakers cannot gauge potential risks to wildlife. Today many fisheries journals demand Bayesian analyses.

Lindley had predicted that the twenty-first century would be a Bayesian era because the superior logic of Bayes' rule would swamp frequency-based methods. David Blackwell at Berkeley disagreed, saying, "If the Bayesian approach does grow in the statistical world, it will not be because of the influence of other statisticians but because of the influence of actuaries, engineers, business people, and others who actually like the Bayesian approach and use it."²⁵ It appeared that Blackwell was right: pragmatism could drive a paradigm shift. Philosophies of science had not changed. The difference was that Bayes finally worked.

Diaconis had been wondering for years, "When is our time?" In 1997, he decided, "Our time is now."²⁶

Smith became the first Bayesian president of the Royal Statistical Society in 1995. Three years later he stunned his friends by quitting statistics to become an administrator of the University of London. A proponent of evidence-based medicine, he wanted to help develop evidence-based public policy too. Dismayed colleagues chastised him for abandoning Bayes' rule. But Smith told Lindley that all the problems of statistics had been solved. We have the paradigm, he said, and with MCMC we know how to implement it. He told Diaconis that there was nothing else to do with statistical problems but to plug them into a computer and turn the Bayesian crank.

In 2008, when Smith became scientific adviser to the United Kingdom's Minister of Innovation, Universities, and Skills, a Royal Society spokesman volunteered that three statisticians have become prime ministers of Great Britain.²⁷

rosetta stones

Two and a half centuries after Bayes and Laplace discovered a way to apply mathematical reasoning to highly uncertain situations, their method has taken wing, soaring through science and the Internet, burrowing into our daily lives, dissolving language barriers, and perhaps even explaining our brains. Gone are the days when a few driven individuals searched orphanages and coded messages for data and organized armies of women and students to make tedious calculations. Today's Bayesians revel in vast archives of Internet data, off-the-shelf software, tools like MCMC, and computing power so cheap it is basically free.

The battle between Bayesian and frequentist forces has cooled. Bayesianism as an all-encompassing framework has been replaced by utilitarian applications and computation. Computer scientists who joined the Bayesian community cared about results, not theory or philosophy. And even theorists who once insisted on adhering strictly to fundamental principles now accept John Tukey's view from the 1950s: "Far better an approximate answer to the right question, . . . than an exact answer to the wrong question." Researchers adopt the approach that best fits their needs.

In this ecumenical atmosphere, two longtime opponents—Bayes' rule and Fisher's likelihood approach—ended their cold war and, in a grand synthesis, supported a revolution in modeling. Many of the newer practical applications of statistical methods are the results of this truce.

As a collection of computational and statistical machinery, Bayes is still driven by Bayes' rule. The word "Bayes" still entails the idea, shared by de Finetti, Ramsey, Savage, and Lindley, that probability is a measure of belief

and that it can, as Lindley put it, “escape from repetition to uniqueness.” That said, most modern Bayesians accept that the frequentism of Fisher, Neyman, and Egon Pearson is still effective for most statistical problems: for simple and standard analyses, for checking how well a hypothesis fits data, and as the foundation of many modern technologies in areas such as machine learning.

Prominent frequentists have also moderated their positions. Bradley Efron, a National Medal of Science recipient who wrote a classic defense of frequentism in 1986, recently told a blogger, “I’ve always been a Bayesian.” Efron, who helped develop empirical Bayesian procedures while remaining a committed frequentist, told me that Bayes is “one of the great branches of statistical inference. . . . Bayesians have gotten more tolerant these days, and frequentists are seeing the need to use Bayesian kinds of reasoning, so maybe we are headed for some kind of convergence.”

Bayes’ rule is influential in ways its pioneers could never have envisioned. “Neither Bayes nor Laplace,” Robert E. Kass of Carnegie Mellon observed, “recognized a fundamental consequence of their approach, that the accumulation of data makes open-minded observers come to agreement and converge on the truth. Harold Jeffreys, the modern founder of Bayesian inference for scientific investigation, did not appreciate its importance for decision-making. And the loyalists of the 1960s and 1970s failed to realize that Bayes would ultimately be accepted, not because of its superior logic, but because probability models are so marvelously adept at mimicking the variation in real-world data.”

Bayes has also broadened to the point where it overlaps computer science, machine learning, and artificial intelligence. It is empowered by techniques developed both by Bayesian enthusiasts during their decades in exile and by agnostics from the recent computer revolution. It allows its users to assess uncertainties when hundreds or thousands of theoretical models are considered; combine imperfect evidence from multiple sources and make compromises between models and data; deal with computationally intensive data analysis and machine learning; and, as if by magic, find patterns or systematic structures deeply hidden within a welter of observations. It has spread far beyond the confines of mathematics and statistics into high finance, astronomy, physics, genetics, imaging and robotics, the military and antiterrorism, Internet communication and commerce, speech recognition, and machine translation. It has even become a guide to new theories about learning and a metaphor for the workings of the human brain.

One of the surprises is that Bayes, as a buzzword, has become chic.

Stanford University biologist Stephen H. Schneider wanted a customized cancer treatment, called his logic Bayesian, got his therapy, went into remission, and wrote a book about the experience. Stephen D. Unwin invented a personal “faith-belief factor” of 28% to boost the 67% “Bayesian probability” that God exists to 95%, and his book hit the bestseller list. A fashionable expression, “We’re all Bayesians now,” plays on comments made years ago by Milton Friedman and President Richard Nixon that “We’re all Keynesians now.” And the CIA agent in a Robert Ludlum thriller tells the hero, “Lucky? Obviously you haven’t heard anything I’ve said. It was a matter of applying Bayes’ Theorem to estimate the conditional probabilities. Giving due weight to the prior probabilities and . . .”¹

It must be conceded that not everyone shares this enthusiasm. Some important fields of endeavor remain opposed. Perhaps the biggest irony is that partisan politics have kept the American census anti-Bayesian, despite Laplace’s vision that enlightened governments would adopt it.

Anglo-American courtrooms are also still largely closed to Bayes. Among the few exceptions was a case in 1994 where Bayes was used to demonstrate that New Jersey state troopers singled out African American drivers for traffic stops. During a rape trial in the 1990s, British lawyers tried teaching judges and juries how to assess evidence using Bayesian probability; judges concluded that the method “plunges the jury into inappropriate and unnecessary realms of theory and complexity.”² Forensic laboratory science in Great Britain and Europe is a different story. Unlike the FBI Laboratory in the United States, the Forensic Sciences Services in Britain have followed Lindley’s advice and now employ Bayesian methods extensively to assess physical evidence. Continental European laboratories have developed a quantitative measure for the value of various types of evidence, much as Turing and Shannon used Bayes to develop bans and bits as units of measurement for cryptography and computers. Bayes—tactfully referred to in forensic circles as the “logical” or “likelihood ratio” approach—has been applied successfully in cases where numbers were available, particularly in DNA profiling. Because DNA databanks involve probabilities about almost unimaginably tiny numbers—one in 20 million, say, or one in a billion—they may eventually open more courtroom doors to Bayesian methods.

Bayes made headlines in 2000 by augmenting DNA evidence with statistical data to conclude that Thomas Jefferson had almost certainly fathered six children by his slave Sally Hemings. DNA evidence from Jefferson’s and Hemings’s families had already offered strong evidence that the third presi-

dent and the author of the Declaration of Independence fathered Hemings's youngest son. But Fraser D. Neiman, the director of archaeology at Jefferson's Monticello plantation, studied whether Hemings's other conceptions fell during or close to one of Jefferson's sporadic visits to Monticello. Then he used Bayes to combine the prior historical testimony and DNA evidence with probable hypotheses based on Jefferson's calendar. Assuming a 50–50 probability that the prior evidence was true, Fraiser concluded it was nearly certain—99% probable—that Jefferson had fathered Hemings's six children.

In economics and finance Bayes appears at multiple levels, ranging from theoretical mathematics and philosophy to nitty-gritty moneymaking. The method figured prominently in three Nobel Prizes awarded for theoretical economics, in 1990, 1994, and 2004. The first Nobel involved the Italian Bayesian de Finetti, who anticipated the Nobel Prize-winning work of Harry Markowitz by more than a decade. Mathematical game theorists John C. Harsanyi and John Nash (the latter the subject of a book and movie, *A Beautiful Mind*) shared a Bayesian Nobel in 1994. Harsanyi often used Bayes to study competitive situations where people have incomplete or uncertain information about each other or about the rules. Harsanyi also showed that Nash's equilibrium for games with incomplete or imperfect information was a form of Bayes' rule.

In 2002 Bayes won perhaps not an entire Nobel Prize but certainly part of one. Psychologists Amos Tversky, who died before the prize was awarded, and Daniel Kahneman showed that people do not make decisions according to rational Bayesian procedures. People answer survey questions depending on their phrasing, and physicians choose surgery or radiation for cancer patients depending on whether the treatments are described in terms of mortality or survival rates. Although Tversky was widely regarded as a philosophical Bayesian, he reported his results using frequentist methods. When James O. Berger of Duke asked him why, Tversky said it was simply a matter of expedience. During the 1970s it was more difficult to publish Bayesian research. "He just took the easy way out," Berger said.

Alan Greenspan, former chairman of the Federal Reserve, said he used Bayesian ideas to assess risk in monetary policy. "In essence, the risk-management approach to monetary policymaking is an application of Bayesian decision-making," Greenspan told the American Economic Association in 2004.³ The audience of academic and government economists gasped; few experts in finance analyze empirical data with Bayes.

Economists were still catching their breaths when Martin Feldstein,

professor of economics at Harvard, stood up at the same meeting and delivered a crash course in Bayesian theory. Feldstein had been President Ronald Reagan's chief economic advisor and was president of the National Bureau of Economic Research, a leading research organization. He learned Bayesian theory at the Howard Raiffa–Robert Schlaifer seminars at Harvard Business School in the 1960s. Feldstein explained that Bayes lets the Federal Reserve weigh a low-probability risk of disaster more heavily than a higher-probability risk that would cause little damage. And he likened Bayes to a man who has to decide whether to carry an umbrella even when the probability of rain is low. If he carries an umbrella but it does not rain, he is inconvenienced. But if he does not carry an umbrella and it pours, he will be drenched. “A good Bayesian,” Feldstein concluded, “finds himself carrying an umbrella on many days when it does not rain.”⁴

Four years later rain flooded the financial markets and banking. Greenspan, who by then had retired from the Federal Reserve, told Congress he had not foreseen the collapse of the real-estate lending bubble in 2008. He did not blame the theory he used but his economic data, which “generally covered only the past two decades, a period of euphoria . . . [instead of] historic periods of stress.”⁵

But did Greenspan actually employ Bayesian statistics to quantify empirical economic data? Or were Bayesian concepts about uncertainty only a handy metaphor? Former Reserve Board governor Alan S. Blinder of Princeton thought the latter, and when he said so during a talk, Greenspan was in the audience and did not object.

In pragmatic contrast to abstract Bayes at the Nobel ceremonies and philosophical Bayes at the Federal Reserve, the rule stands behind one of the most successful hedge funds in the United States. In 1993 Renaissance Technologies hired away from IBM a Bayesian group of voice recognition researchers led by Peter F. Brown and Robert L. Mercer. They became co-managers of RenTech's portfolio and technical trading. For several years, their Medallion Fund, limited to former and current employees, averaged annual returns of about 35%. The fund bought and sold shares so rapidly one day in 1997 that it accounted for more than 10% of all NASDAQ trades.

To search for the nonrandom patterns and movements that will help predict markets, RenTech gathers as much information as possible. It begins with prior knowledge about the history of prices and how they fluctuate and correlate with each other. Then the company continuously updates that prior base. As Mercer explained, “RenTec gets a trillion bytes of data a day, from

newspapers, AP wire, all the trades, quotes, weather reports, energy reports, government reports, all with the goal of trying to figure out what's going to be the price of something or other at every point in the future. . . . We want to know in three seconds, three days, three weeks, three months. . . . The information we have today is a garbled version of what the price is going to be next week. People don't really grasp how noisy the market is. It's very hard to find information, but it is there, and in some cases it's been there for a long, long time. It's very close to science's needle-in-a-haystack problem."

Like investors at RenTech, astronomers, physicists, and geneticists use Bayes to discern elusive phenomena almost drowning in unknowns. A scientist may face hundreds of thousands of variables without knowing which produce the best predictions. Bayes lets them estimate the most probable values of their unknowns.

When Supernova 1987A exploded, astronomers detected precisely 18 neutrinos. The particles originated from deep inside the star and were the only clues about its interior, so the astronomers wanted to extract as much information as possible from this minuscule amount of data. Tom Loredo, a graduate student at the University of Chicago, was told to see what he could learn. Because the supernova was a one-of-a-kind opportunity, frequency-based methods did not apply. Loredo began reading papers by Lindley, Jim Berger, and other leading Bayesians and discovered that Bayes would let him compare various hypotheses about his observations and choose the most probable. His Ph.D. thesis from 1990 wound up introducing modern Bayesian methods to astronomy.

Since then, Bayes has found a comfortable niche in high-energy astrophysics, x-ray astronomy, gamma ray astronomy, cosmic ray astronomy, neutrino astrophysics, and image analysis. In physics, Bayes is hunting for elusive neutrinos, Higgs-Boson particles, and top quarks. All these problems deal with needles in haystacks, and Loredo now uses Bayes at Cornell University in a new field, astrostatistics.

In much the same way, biologists who study genetic variation are limited to tiny snippets of information almost lost among huge amounts of meaningless and highly variable data in the chromosomes. Computational biologists searching for genetic patterns, leitmotifs, markers, and disease-causing misspellings must extract the weak but important signals from the deafening background noise that masquerades as information.

Susan Holmes, a professor in Stanford's statistics department, works in computational and molecular biology on amino acids. Some are extremely

rare, and if she used frequentist methods she would have to quantify them with a zero. Adopting the cryptographic technique used by Turing and Good at Bletchley Park, she tries to crack the genetic code by assigning missing species a small possibility.

Given that the DNA in every biological cell contains complete instructions for making every kind of protein in the body, what differentiates a kidney cell from a brain cell? The answer depends on whether a particular gene is turned on or off and whether the genes work together or not. Holmes assembles huge microarrays of genetic data filled with noise and other distractions that may hide a few important signals from the turned-on genes. Each microarray consists of many genes arrayed in a regular pattern on a small glass slide or membrane; with it, she can analyze the expression of thousands of genes at once.

"It's very tenuous," she says. "[Imagine that] you have a city at night like Toronto or Paris with a very dense population and lots of buildings, and at 2 a.m. you look at which lights are lit up in all the buildings. Then at 3 and 4 a.m., you look again. So you develop a pattern of which rooms are lit up, and from that you infer who in the city knows who. That's how sparse the signal is and how far you have to jump, to see which genes are working together. You don't even have phone connections. But the image of something lighting up is a little bit like the image of microarrays. Microarrays have so much noise, it seems crazy. You just have rustles, whispers of signals, and then lots of noise. You spend a lot of time looking at a lot of data." Because prior information is needed to assemble the networks, many microarrays are analyzed using Bayesian methods.

Daphne Koller, a leader in artificial intelligence and computational biology at Stanford, also works on microarrays. She wanted to see not only which genes have turned on or off, but also what controls and regulates them. By looking at the activity levels of genes in yeast, she figured out how they are regulated. Then she switched to mouse and human cells to determine the differences in genetic regulation between healthy people and patients with cancer or Type II diabetes, particularly the metabolic (insulin resistance) syndrome.

On the vexed issue of priors, Koller considers herself a relaxed middle-of-the-roader. In contrast, Bayesian purists like Michael I. Jordan of Berkeley and Philip Dawid of Cambridge object to the term "Bayesian networks"; they regard Judea Pearl's nomenclature as a misnomer because Bayesian networks do not always have priors and Bayes without priors is not Bayes. But Koller

insists that her networks fully qualify as Bayesian because she carefully constructs priors for their variables.

Koller's fascination with uncertainty has led her from genetics to imaging and robotics. Images typically have variable and ambiguous features that are embedded in background clutter. The human visual system sends ten million signals per second to the brain, where a billion neurons strip off random fluctuations and irrelevant, ambiguous information to reveal shape, color, texture, shading, surface reflections, roughness, and other features. As a result, human beings can look at a blurry, distorted, noisy pattern and instantly recognize a tomato plant, a car, or a sheep. Yet a state-of-the-art computer trained to recognize cars and sheep may picture only nonsensical rectangles. The difference is that the human brain has prior knowledge to integrate with the new images.

"It's mind-boggling," says Koller. The problem is not computer hardware; it is writing the software. "A computer can easily be trained to distinguish a desert from a forest, but where the road is and where it's about to fall off a cliff, that's much harder."

To explore such imaging problems, Sebastian Thrun of Stanford built a driverless car named Stanley. The Defense Advanced Research Projects Agency (DARPA) staged a contest with a \$2 million prize for the best driverless car; the military wants to employ robots instead of manned vehicles in combat. In a watershed for robotics, Stanley won the competition in 2005 by crossing 132 miles of Nevada desert in seven hours.

While Stanley cruised along at 35 mph, its camera took images of the route and its computer estimated the probability of various obstacles. As the robot navigated sharp turns and cliffs and generally stayed on course, its computer could estimate with 90% probability that a wall stood nearby and with a 10% probability that a deep ditch was adjacent. In the unlikely event that Stanley fell into the ditch, it would probably have been destroyed. Therefore, like the Bayesian economist who carries an umbrella on sunny days, Stanley slowed down to avoid even unlikely catastrophes. Thrun's artificial intelligence team trained Stanley's sensors, machine-learning algorithms, and custom-written software in desert and mountain passes.

Thrun credited Stanley's victory to Kalman filters. "Every bolt of that car was Bayesian," Diaconis said proudly. After the race, Stanley retired in glory to its own room in the Smithsonian National Museum of American History in Washington.

The next year a Bayesian team from Carnegie Mellon University and

General Motors won another \$2 million from DARPA by maneuvering a robot through city traffic while safely avoiding other cars and obeying traffic regulations. Urban planners hope driverless cars can solve traffic congestion. Another Carnegie Mellon team relied on Bayes' rule and Kalman filters to win international robotic soccer championships involving fast-moving multirobot systems.

The U.S. military is heavily involved in imaging issues. Its Automatic Target Recognition (ATR) technology is a heavy user of Bayesian methods for robotic and electronic warfare, combat vehicles, cruise missiles, advanced avionics, smart weapons, and intelligence, surveillance, and reconnaissance. ATR systems employ radar, satellites, and other sensors to distinguish between, for example, a civilian truck and a missile launcher. Some ATR computer programs start with Bayes' controversial 50–50 odds, even though these can have a strong impact on rare events and better information may be available. Echoing generations of critics, at least one anonymous ATR analyst regards Bayes as “an affront, a cheap easy trick. It depends on an initial hunch. And yet it turns out to be an effective approximation that seems to solve many of the world's problems. So Bayes' rule is wrong . . . except for the fact that it works.” Other approaches have been computationally more expensive and did not produce better answers.

Besides imaging problems, the military involves Bayes in tracking, weapons testing, and antiterrorism. Reagan's Ballistic Missile Defense applied a Bayesian approach to tracking incoming enemy ballistic missiles. Once it was sufficiently probable that a real missile had been detected, Bayes allowed sensors to communicate only their very latest data instead of recalculating an entire problem from scratch each time. The National Research Council of the National Academy of Sciences strongly urged the U.S. Army to use Bayesian methods for testing weapons systems, specifically the Stryker family of light, armored vehicles. Many military systems cannot be tested in the large sample sizes required by frequentist methods. A Bayesian approach allows analysts to combine test data with information from similar systems and components and from earlier developmental tests. Terrorist threats are generally estimated with Bayesian techniques. Even before the attacks of September 11, 2001, Digital Sandbox of Tyson's Corner, Virginia, used Bayesian networks to identify the Pentagon as a possible target. Bayes combined expert and subjective opinions about possible events that had never occurred.

The United States is not the only country trying to predict terrorism. As Britain considered building a national data bank to detect potential terrorists,

Bayes raised the same alarm it had against mass HIV screening. Terrorists are so rare that the definition of a terrorist will have to be extremely accurate or else many, many people will be identified as dangerous when they are, in fact, not at all.

On the Internet Bayes has worked its way into the very fiber of modern life. It helps to filter out spam; sell songs, books, and films; search for web sites; translate foreign languages; and recognize spoken words. David Heckerman, who used Bayesian networks to diagnose lymph node diseases for his Ph.D. thesis, has the modern practitioner's wide-open attitude about Bayes: "The whole thing about being a Bayesian is that all probability represents uncertainty and, anytime you see uncertainty, you represent it with probability. And that's a lot bigger than Bayes' theorem."

In 1992 Heckerman moved from Stanford to Microsoft, where he founded and manages the Machine Learning and Applied Statistics Group of Microsoft Research. The problems there are very different. Because Stanford had plenty of experts but little data, he says he built Bayesian nets with priors based on expert opinion: "But Microsoft had lots of data and only a few experts, so we got into combining expert knowledge with data." One of Microsoft's first Bayesian applications helped parents with sick children type in their children's symptoms and learn the best course of action. In 1996 Bill Gates, cofounder of Microsoft, made Bayes headline news by announcing that Microsoft's competitive advantage lay in its expertise in Bayesian networks.

That same year Heckerman, Robert Rounthwaite, Joshua Goodman, Eric Horwitz, and others began investigating Bayesian antispam techniques. Remember vVi-@-gra, l0w m0rtg@ge rates, PARTNERSHIP INVESTMENT, and !!!! PharammcyByMAIL? Advertisements that are unwanted and often pornographic and fraudulent are sent to millions without their permission. Spam soon accounted for more than half of all mail on the Internet, and some e-mail users spent half an hour a day weeding it out.

Bayesian methods attack spam by using words and phrases in the message to determine the probability that the message is unwanted. An e-mail's spam score can soar near certainty, 0.9999, when it contains phrases like "our price" and "most trusted"; coded words like "genierc virgaa"; and uppercase letters and punctuation like !!! or \$\$\$\$. High-scoring messages are automatically banished to junk mail files. Users refine their own filters by reading low-scoring messages and either keeping them or sending them to trash and junk files. This use of Bayesian optimal classifiers is similar to the technique

used by Frederick Mosteller and David Wallace to determine who wrote certain *Federalist* papers.

Bayesian theory is firmly embedded in Microsoft's Windows operating system. In addition, a variety of Bayesian techniques are involved in Microsoft's handwriting recognition; recommender systems; the question-answering box in the upper right corner of a PC's monitor screen; a data-mining software package for tracking business sales; a program that infers the applications that users will want and preloads them before they are requested; and software to make traffic jam predictions for drivers to check before their commute.

Bayes was blamed—unfairly, say Heckerman and Horwitz—for Microsoft's memorably annoying paperclip, Clippy. The cartoon character was originally programmed using Bayesian belief networks to make inferences about what a user knew and did not know about letter writing. After the writer reached a certain threshold of ignorance and frustration, Clippy popped up cheerily with the grammatically improper observation, "It looks like you're writing a letter. Would you like help?" Before Clippy was introduced to the world, however, non-Bayesians had substituted a cruder algorithm that made Clippy pop up irritatingly often. The program was so unpopular it was retired.

Bayes and Laplace would probably be appalled to learn that their work is heavily involved in selling products. Much online commerce relies on recommender filters, also called collaborative filters, built on the assumption that people who agreed about one product will probably agree on another. As the e-commerce refrain goes, "If you liked this book/song/movie, you'll like that one too." The updating used in machine learning does not necessarily follow Bayes' theorem formally but "shares its perspective." A \$1-million contest sponsored by Netflix.com illustrates the prominent role of Bayesian concepts in modern e-commerce and learning theory. In 2006 the online film-rental company launched a search for the best recommender system to improve its own algorithm. More than 50,000 contestants from 186 countries vied over the four years of the competition. The AT&T Labs team organized around Yehuda Koren, Christopher T. Volinsky, and Robert M. Bell won the prize in September 2009.

Interestingly, although no contestants questioned Bayes as a legitimate method, almost none wrote a formal Bayesian model. The winning group relied on empirical Bayes but estimated the initial priors according to their

frequencies. The film-rental company's data set was too big and too filled with unknowns for anyone to—almost instantaneously—create a model, assign priors, update posteriors repeatedly, and recommend films to clients. Instead, the winning algorithm had a Bayesian “perspective” and was laced with Bayesian “flavors.” However, by far the most important lesson learned from the Netflix competition originated as a Bayesian idea: sharing.

Volinsky had used Bayesian model averaging for sharing and averaging complementary models while working in 1997 on his Ph.D. thesis about predicting the probability that a patient will have a stroke. But the Volinsky and Bell team did not employ the method directly for Netflix. Nevertheless, Volinsky emphasized how “due to my Bayesian Model Averaging training, it was quite intuitive for me that combining models was going to be the best way to improve predictive performance. Bayesian Model Averaging studies show that when two models that are not highly correlated are combined in a smart way, the combination often does better than either individual model.” The contest publicized Bayes' reputation as a fertile approach to learning far beyond mere Bayesian technology.

Web users employ several forms of Bayes to search through the billions of documents and locate what they want. Before that can happen, though, each document must be profiled or categorized, organized, and sorted, and its probable interconnectedness with other documents must be calculated. At that point, we can type into a search engine the unrelated keywords we want to appear in a document, for example, “parrots,” “madrigals,” and “Afghan language.” Bayes' rule can winnow through billions of web pages and find two relevant ones in 0.31 seconds. “They're inferential problems,” says Peter Hoff at the University of Washington. “Given that you find one document interesting, can you find other documents that will interest you too?”

When Google starts projects involving large amounts of data, its giant search engines often try naïve Bayesian methods first. Naïve Bayes assumes simplistically that every variable is independent of the others; thus, a patient's fever and elevated white blood cell counts are treated as if they had nothing to do with each other. According to Google's research director, Peter Norvig, “There must have been dozens of times when a project started with naïve Bayes, just because it was easy to do and we expected to replace it with something more sophisticated later, but in the end the vast amount of data meant that a more complex technique was not needed.”

Google also uses Bayesian techniques to classify spam and pornography

and to find related words, phrases, and documents. A very large Bayesian network finds synonyms of words and phrases. Instead of downloading dictionaries for a spelling checker, Google conducted a full-text search of the entire Internet looking for all the different ways words can be spelled. The result was a flexible system that could recognize that “shaorn” should have been “Sharon” and correct the typo.

While Bayes has helped revolutionize modern life on the web, it is also helping to finesse the Tower of Babel that has separated linguistic communities for millennia. During the Second World War, Warren Weaver of the Rockefeller Foundation was impressed with how “a multiplicity of languages impedes cultural interchange between the peoples of the earth and is a serious deterrent to international understanding.”⁶ Struck by the power of mechanized cryptography and by Claude Shannon’s new information theory, Weaver suggested that computerized statistical methods could treat translation as a cryptography problem. In the absence of computer power and a wealth of machine-readable text, Weaver’s idea lay fallow for decades.

Ever since, the holy grail of translators has been a universal machine that can transform written and spoken words from one language into any other. As part of this endeavor, linguists like Noam Chomsky developed structural rules for English sentences, subjects, verbs, adjectives, and grammar but failed to produce an algorithm that could explain why one string of words makes an English sentence while another string does not.

During the 1970s IBM had two competing teams working on a similar problem, speech recognition. One group, filled with linguists, studied the rules of grammar. The other group, led by Mercer and Brown, who later went to RenTech, was filled with mathematically inclined communications specialists, computer scientists, and engineers. They took a different tack, replaced logical grammar with Bayes’ rule, and were ignored for a decade.

Mercer’s ambition was to make computers do intelligent things, and voice recognition seemed to be the way to make this happen. For both Mercer and Brown speech recognition was a problem about taking a signal that had passed through a noisy channel like a telephone and then determining the most probable sentence that the speaker had in mind. Ignoring grammar rules, they decided to figure out the statistical probability that words and phrases in one language would wind up as particular words or phrases in another. They did not have to know any language at all. They were simply computing the probability of a single word given all the words that preceded

it in a sentence. For example, by looking at pairs of English words they realized that the word after “the” was highly unlikely to be “the” or “a,” somewhat more likely to be “cantaloupe,” and still more likely to be “tree.”

“It all hinged on Bayes’ theorem,” Mercer recalled. “We were given an acoustic output, and we’d like to find the most probable word sequence, given the acoustic sequence that we heard.” Their prior knowledge consisted of the most probable order of the words in an English sentence, which they could get by studying enormous amounts of English text.

The big problem throughout the 1970s was finding enough data. They needed bodies of text focused on a fairly small topic, but nothing as adult as the *New York Times*. At first they worked their way through old, out-of-copyright children’s books; 1,000 words from a U.S. Patent Office experiment with laser technology; and 60 million words of Braille-readable text from the American Printing House for the Blind.

At an international acoustic, speech, and signal meeting the group wore identical T-shirts printed with the words “Fundamental Equation of Speech Recognition” followed by Bayes’ theorem. They developed “a bit of swagger, I’m ashamed to say,” Mercer recalled. “We were an obnoxious bunch back in those days.”

In a major breakthrough in the late 1980s they gained access to French and English translations of the Canadian parliament’s daily debates, about 100 million words in computer-readable form. From them, IBM extracted about three million pairs of sentences, almost 99% of which were actual translations of one another. It was a Rosetta Stone in English and French. “You had a day’s worth of English and the corresponding day’s worth of French, so things were lined up to that extent, but we didn’t know that this sentence or word went with this or that sentence or word. For example, while the English shouted, ‘Hear! Hear!’ the French said, ‘Bravo!’ So we began working on getting a better alignment of the sentences. We were using the same methods as in speech recognition: Bayes’ theorem and hidden Markov models.” The latter are particularly useful for recognizing patterns that involve likely time sequences, for example, predicting one word in a sentence based on the previous word.

In a landmark paper in 1990 the group applied Bayes’ theorem to full sentences. There was a small probability that the sentence, “President Lincoln was a good lawyer,” means “Le matin je me brosse les dents” but a relatively large probability that it means “Le président Lincoln était un bon avocat.” After that paper, several of the leading machine translation systems incorporated Bayes’ rule.

In 1993, lured by lucre and the challenge, Mercer and Brown moved from IBM and machine translation to RenTech, where they became vice presidents and co-portfolio managers for technical trading. So many members of IBM's speech recognition group joined them that critics complain they set back the field of machine translation five years.

After the 9/11 disaster and the start of the war in Iraq, the military and the intelligence communities poured money into machine translation. DARPA, the U.S. Air Force, and the intelligence services want to ease the burden on human translators working with such little-studied languages as Uzbek, Pashto, Dari, and Nepali.

Machine translation got still another boost when Google trawled the Internet for more Rosetta Stone texts: news stories and documents published in both English and another language. United Nations documents alone contributed 200 billion words. By this time the web was churning out enormous amounts of text, free for the asking. Combing English words on the web, Google counted all the times that, for example, a two-word sequence in English meant "of the." To determine which words in the English sentence correspond to which words in the other language, Google uses Bayes to align the sentences in the most probable fit.

The blue ribbons Google won in 2005 in a machine language contest sponsored by the National Institute of Standards and Technology showed that progress was coming, not from better algorithms, but from more training data. Computers don't "understand" anything, but they do recognize patterns. By 2009 Google was providing online translations in dozens of languages, including English, Albanian, Arabic, Bulgarian, Catalan, Chinese, Croatian, Czech, Danish, Dutch, Estonian, Filipino, Finnish, and French.

The Tower of Babel is crumbling.

Even as Bayes' rule was improving human communications, it was returning full circle to the fundamental question that had occupied Bayes, Price, and Laplace: How do we learn? Using Bayes' rule, more than half a million students in the United States learn the answer to that question each year: we combine old knowledge with new. Approximately 2,600 secondary schools teach algebra and geometry with Bayesian computer programs developed at Carnegie Mellon University since the late 1980s. The software also teaches French, English as a second language, chemistry, physics, and statistics.

The programs, called Cognitive Tutors, are based on John R. Anderson's idea that Bayes is a surrogate for the way we naturally learn, that is to say,

gradually. The ability to accumulate evidence is an optimal survival strategy, but our brains cannot assign a high priority to everything. Therefore, most students must see and work many times with a mathematical concept before they can retrieve and apply it at will. Our ability to do so depends on how frequently and recently we studied the concept.

In addition to viewing Bayes as a continuous learning process, Cognitive Tutors depend on Bayes' theorem for calculating each student's "skillometer," the probability that the individual has mastered a topic and is ready for a new challenge. Ten years after this double-edged Bayesian approach was launched, its students were learning as much or more than conventionally taught pupils—in a third the time.

The flowering of Bayesian networks, neural nets, and artificial intelligence nets has helped neuroscientists study how the brain's neurons process information that arrives directly and indirectly, a little at a time, in tiny, often contradictory packets. As a computational tool and learning theory, Bayes has been involved in mapping the brain and analyzing its circuitry as well as in decoding signals from neurons and harnessing them to make better prostheses and robots.

Hundreds of megabits of sensory information bombard the waking brain every second. From that stream of data, 10 billion nerve cells extract information and correct prior understanding several times every 100 milliseconds. Discerning which sensory stimulus has caused which neuronal response is a difficult problem: the neurons fire unpredictably, scientists cannot monitor all of them at once, and the brain combines cues from multiple sources. The vision regions of our brains, for example, create three-dimensional objects and scenes. To do so, they rely on our prior knowledge about the regularities in our environment—for example, that light generally shines from above and that straight lines and 90-degree angles are apt to be man-made. But our brains refine that knowledge with new data pouring in about depth, contours, symmetry, lines of curvature, texture, shading, reflectance, foreshortening, and motion.

In 1998 the neurostatistician Emery N. Brown of MIT and Massachusetts General Hospital realized that Bayesian methods might deal with these uncertainties. Using Kalman filters, he and the MIT neuroscientist Matthew A. Wilson described a rat's brain as it processed information about the animal's location in its environment. Approximately 30 so-called place neurons in the hippocampus keep a rat informed about its location. As a laboratory rat for-

aged in a box scattered randomly with chocolate tidbits, electrodes implanted in its brain imaged some of the place neurons as they fired. A Bayesian filter sequentially updated the rat's position in the box. The researchers could see neither the rat nor its box, but by watching the neurons fire they could track the rat's movements. Thanks to Bayes, Brown could reconstruct the path of the chocolate-loving rat with only a fifth or a tenth of the neurons that previous methods had required.

To explore the practicality of using the living brain to power prostheses and robots, Brown's statistical method was replicated with a few dozen motor neurons, Bayesian algorithms, and Bayesian particle filters. The goal is to develop an artificial arm that can smoothly reach, rotate the hand, move fingers independently, and grasp and retrieve objects. Illustrating the possibilities of the approach, a rhesus monkey in Andrew B. Schwartz's laboratory at the University of Pittsburgh gazed longingly at an enticing treat. With his arms immobilized in plastic tubes and his mouth salivating, the motor neurons in his brain fired repeatedly, activating a robotic arm. The monkey's control was so precise it could reach out with the robotic arm, nab the treat, and fetch it back to eat. Frequentist methods can deal with simple backwards-and-forwards motion, but Bayesian neurostatisticians believe their algorithms will be powerful and flexible enough to control a robotic arm's position, rotation, acceleration, velocity, momentum, and grasp.

These attempts to capitalize on all the information available in neurons raise questions: What does the brain itself do? Does it maximize the information it gets from an uncertain world by performing Bayesian-like computations? In discussions of these questions, Bayes has become more than just an aid for data analysis and decision making. It has become a theoretical framework for explaining how the brain works. In fact, as such, the "Bayesian Brain" has become a metaphor for a human brain that mimics probability.

In our struggle to survive in an uncertain and changing world, our sensory and motor systems often produce signals that are incomplete, ambiguous, variable, or corrupted by random fluctuations. If we put one hand under a table and estimate its location, we can be off by up to 10 centimeters. Every time the brain generates a command for action, we produce a slightly different movement. In this confusing world, Bayes has emerged as a useful theoretical framework. It helps explain how the brain may learn. And it demonstrates mathematically how we combine two kinds of information: our prior beliefs about the world with the error-fraught evidence from our senses.

As Lindley emphasized years ago, if we are certain about the evidence

relayed by our senses, we rely on them. But when faced with unreliable sensory data, we fall back on our prior accumulation of beliefs about the world.

When Daniel Wolpert of Cambridge University tested the theory with a virtual tennis game, he showed that players unconsciously combine their prior knowledge of bouncing balls in general with sensory data about a particular ball coming over the net. In short, they unconsciously behave like good Bayesians. In addition, Wolpert said, the nice thing about Bayes was that it did not produce a single number. It made multiple predictions about every possible state given the sensory data. Thus, the tennis ball would most probably bounce in a particular spot—but there was also a reasonable chance it would fall elsewhere.

According to Bayes, the brain stores a wide range of possibilities but assigns them high and low probabilities. Color vision is already known to operate this way. We think we perceive red, but we actually see an entire spectrum of colors, assign the highest probability to red, **and** keep in mind the outside possibilities that the color could be pink or purple.

Wolpert concluded that Bayesian thinking is basic to everything a human does, from speaking to acting. The biological brain has evolved to minimize the world's uncertainties by thinking in a Bayesian way. In short, growing evidence suggests that we have Bayesian brains.

Given Bayes' contentious past and prolific contributions, what will the future look like? The approach has already proved its worth by advancing science and technology from high finance to e-commerce, from sociology to machine learning, and from astronomy to neurophysiology. It is the fundamental expression for how we think and view our world. Its mathematical simplicity and elegance continue to capture the imagination of its users.

But what about the years to come? Brute computer force can organize stunning quantities of information, but it clusters and searches for documents crudely, according to keywords. Only the brain examines documents and images according to their meaning and their content. Which approach will be more useful? Will computers become so powerful that huge amounts of data alone will teach us everything? Will scientists no longer need to theorize or hypothesize before experimenting or gathering their data? Or will Bayesian organizational principles remain fundamental? Current strategies for designing computers that could perform at biological levels exploit such ancient principles as reusable parts, hierarchical structures, variations on themes, and regulatory systems.

The jumping off point for this debate is Bayes and its priors, says Stuart Geman, whose Gibbs sampler helped launch the modern Bayesian revolution: “In this debate, there is no more powerful argument for Bayes than its recognition of the brain’s inner structures and prior expectations.” The old controversies between Bayesians and frequentists have been reframed in terms of, Do we use probabilities or not? Old or new, the issues are similar, if not identical, Geman says. And in its new guise, Bayesian learning and its priors occupy the heart of the debate.

Can we look forward to a time when computers can compete with our biological brains for understanding? Will they be programmed with Bayes? Or with something else?

Whatever the outcome of the revolution, Diaconis insists that Bayes will play a role. “Bayes is still young. Probability did not have any mathematics in it until 1700. Bayes grew up in data-poor and computationally poor circumstances. It hasn’t settled down yet. We can give it time.

“We’re just starting.”

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